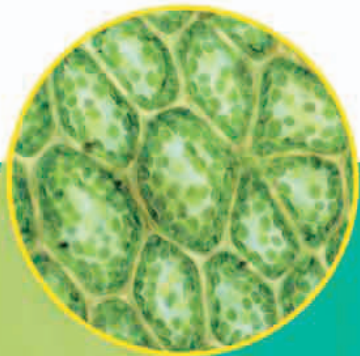


PHOTOSYNTHESIS

Seaweeds as well as seagrass and phytoplankton make food by converting carbon dioxide and water into oxygen and sugar. This process is called photosynthesis. It happens inside microscopic structures called chloroplasts. These contain a green substance called chlorophyll that can absorb solar energy. The energy triggers the chemical reaction that makes the sugar, so if there is not enough light, photosynthesis cannot take place.



▲ SUGAR FACTORIES

The fronds of big seaweeds are made up of millions of microscopic cells linked together like bricks in a wall. Each cell contains many green chloroplasts that use light to make sugar compounds.

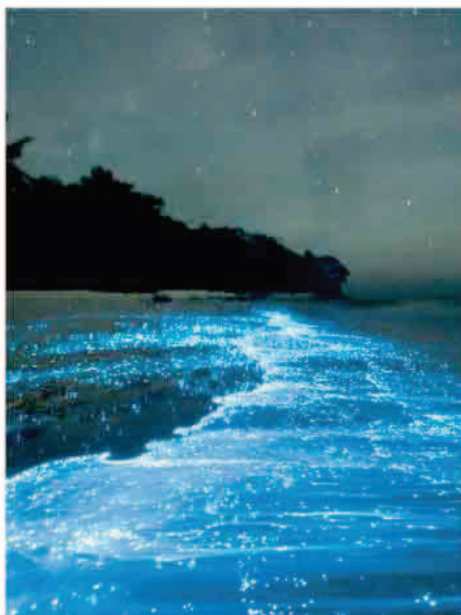
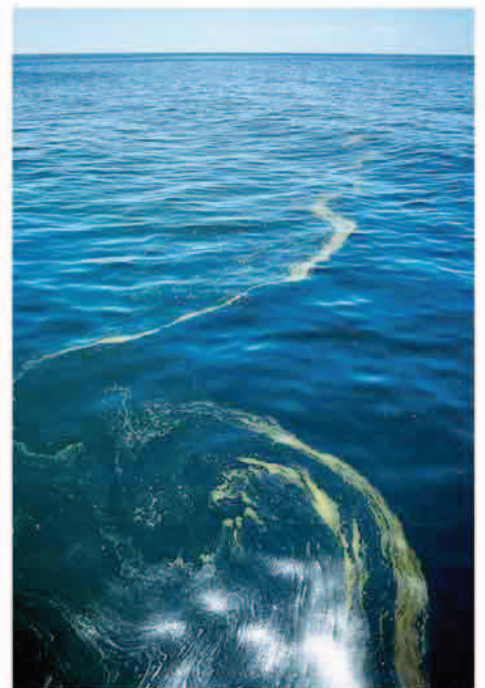


SUNLIT DRIFTERS

Like seaweeds, the tiny drifting organisms of the phytoplankton can live only in the upper, sunlit zone of the ocean, where there is enough light for them to make sugar. But unlike seaweeds, they all consist of just one living cell. They range from very simple bacteria to complex single-celled algae, including diatoms, dinoflagellates, and coccolithophores. Diatoms have intricate skeletons made of glassy silica; seen here under a microscope, they look like miniature jewelry. Coccolithophores have similar skeletons made of chalky calcite.

BLOOMING OCEANS

Although the individual organisms that make up phytoplankton are only microscopic, they can form dense blooms near the ocean surface that are sometimes visible from space. These usually develop where the sea is rich in minerals, drawn up from deeper water by ocean currents. The diatoms and other organisms absorb these minerals and use them to make their skeletons. They also combine them with sugar compounds to make other substances vital to their survival.



LIVING LIGHT

Since phytoplankton are too small to be seen without a microscope, we normally see the organisms as a cloudiness in the water. The richer the sea, the more phytoplankton it can support, and the cloudier it is. But some of these drifting organisms, such as certain types of dinoflagellate, can glow with a blue-green chemical light when they are disturbed. This creates dazzling effects at night, as seen on this tropical shore.

